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Education

California State University, Fullerton - 2024 thru now

- Pursuing a MS in both Computer and Civil Engineering
- Having a solid foundation in how software works at a high level serves me for certain purposes:
- Computer Engineering*: Learning about how to manipulate hardware helps me solve even more pressing technical issues.
- Civil Engineering*: The field pays well. It also helps to learn these disparate industries well enough to know how to bridge them and make good for myself and society.

University of California, Irvine - 2020 thru 2024

- Completed a BS in Software Engineering with a GPA of 3.671.
- For a year in 2022, I participated at ACM club meetings,
- Verbalized technical problem-solving approaches for *easy* and *medium* LeetCode problems to an audience of various skill levels.

Santiago Canyon College - 2018 thru 2020

- Majoring in Computer Science in preparation for a transfer to UCI; graduated with an associate's degree in Liberal Arts.
- Achieved a 4.0 GPA for major-related courses.

Listed Skills

- Python
- C++
- Rust
- JavaScript
- HTML5; CSS3
- PHP
- Roblox Lua(u)
- x86 Assembly

- MongoDB
- MySQL
- Next.js; React
- Docker Compose
- Git; GitHub
- GNU/Linux
- Bash
- PowerShell
- Android/Linux
- FFmpeg

- Affinity Photo
- FL Studio
- OBS Studio
- Filmora 13
- VS Code; VSCodium
- Documenting Stuff
- Microsoft Office

Employment

Engineering Assistant @ Professional Engineering Center - June 2015 thru now

- Safely managing mission-critical on-site SMB-compatible file servers with sensitive data going back since 2003, enabling reliable file access for the engineering team.
- Translating hand-drawn sketches into production-ready AutoCAD drawings for licensed engineer review and sign-off, accelerating project turnaround.
- Directly contributed to the organisational design of high-stakes commercial and industrial projects for multi-national manufacturers and regional utility providers.
- Reverse-engineered and modified legacy plugins to fully support modern versions of AutoCAD, enabling full migration to an updated toolchain and reducing operational risk.

Raising Cane's and El Pollo Loco - Aug 2019 thru June 2021

- Communicated with customers and team members in a fast-paced environment.
- Operated drive-thru and dine-in cash registers with total revenue exceeding \$3000 per day; processing orders through the POS system and supporting overall restaurant operations.
- Helped maintain steady order throughput during peak hours.

• On-call Designer @ Islamic Institute of Orange County et al. - April 2016 thru now

- Controlling graphic design for printed and digital materials for social events leveraging Affinity Photo and Canva, ensuring brand consistency and on-time delivery.
- Creating marketing copy for affiliated non-profits such as Sabil USA, BSA Scouts, and Girl Scouts of America.
- Matching design language with existing literature and coordinating with print shops, reducing revision.
- Additionally, designed real-estate visuals for agents contracted by Berkshire Hathaway.

• Research Internship @ CSU Fullerton - Sep 2017 thru May 2018

- Analysed *my first* proper code base and documented findings that helped the research team streamline future development.
- Supported graduate students in building a Java-based brain-computer interface targeting users with limited mobility.
- Coordinated with Professor Kiran George under the Val Tech Program to log internship hours, ensuring that I and the team met our reporting requirements.
- My assigned task:* To design a mock user interface for initial calibration procedure.
- Salient Moment:** I identified and took the initiative to solve issues which were outside my purview.
- During a meeting, I overheard that the researchers were facing challenges in accessing a specific data structure obscured by dependency code, which blocked their analysis.
- I located the data structure in a private variable buried in a library class.
- To add some spice, I drafted a digital illustration showing the variable's name and presented that to the group.
- However, my solution was not actionable as it required changes to the dependency code and overlooked other project needs.

Projects & Research

• Rōblox Freedom Distribution - July 2023 thru now

- <https://github.com/Windows81/Roblox-Freedom-Distribution>
- I have been building games on Rōblox from 2013, having had an account since 2008.
- At the time, the company was a small Silicon-Valley start-up.
- Over the years, its developer-friendly tools led me to take up my current-day career as a developer.
- From 2023, I've built a scalable application for users to *host* and *join* self-hosted, historic versions of Rōblox in a local- or wide-area network.
- Created a Python bootstrapper, dually for end users and custom backends, to enable seamless connectivity.
- Enhanced performance through team collaboration, lots and lots of code reviews, and feature development.
- Consulted cutting-edge academic papers to validate my Python re-write of Edgebreaker geometry-compression.
- Built a automation asset-processing pipeline for proprietary 3-dimensional mesh and rbxl formats, promoting quality.
- Evaluated fair-use compliance by auditing business-legal literature, publicly disclosing some analysis of legal risks and benefits of various intended use cases.

• Front-end | Frontend Developer at Gameln - Sep 2023 thru Mar 2024

- Before graduating from UC Irvine, I took a capstone course where I collaborated, among classmates, with one of a choice of several companies.
- Owing to my prior experience in videogame development, I chose to work with Gameln, who were building a tool to match e-sports streamers and paying sponsors.
- Implemented mobile-user-friendly pages; leveraged Next.js (and thus React) from mockups that we also made.
- Arranged, as part of a team, a slide deck with stylistic help from Figma.
- The slide deck was presented to an audience of approx. 50 other students and their families.

• Bring Back Buffets - March & April 2021

- <https://github.com/Windows81/Bring-Back-Bufferets>
- During the Covid lockdowns, I engineered a mobile web app to revive buffets whilst minimising infection risk.
- This project's user-interaction model drew inspiration from how some Korean barbecue establishments operate:
 - Where dine-in customers can order food directly from a website they can access from their phones.
- I built separate user interfaces for customers, cashiers, and back-of-house staff, with step-by-step documentation for setup.
- Although the project is fully developed, I was unable to secure clients.

• GoAnimate Wrapper - Dec 2019 thru Nov 2020

- <https://github.com/GoAnimate-Wrapper/GoAnimate-Wrapper>
- Using GoAnimate's *Legacy Video Maker*, I animated some engaging videos on my YouTube channel, often satirising trends set forth by other publishers who use the same tool.
- However, it needed a remote server for file persistence and access to crucial assets, which was being discontinued after 2019.
- A newer version of the program existed, but lacked crucial animation assets and effects.
- GoAnimate Wrapper is a server-backend application designed to interop with Vyond's Legacy Video Maker.
- Reverse-engineered Flash files and analysed legacy ActionScript code.
- Created and scaled a modern JavaScript interface replicating GoAnimate's back-end functionalities.
- With a team of experts managing the community up to this day, Wrapper became my most popular application.
- Because of the need that it filled, my tool delivered hundreds of engaging video experiences.
- By May 2021, over 21,000 results would appear on a YouTube query for Wrapper and its successors.

• "Screwaja YuJa" - Dec 2022

- <https://github.com/Windows81/Screwaja-YuJa>
- I discovered and disclosed a severe vulnerability in YuJa's API.
- Over 6,000,000 academic videos, including private recordings of students taking proctored exams, were exposed on the open net.
- After sending the initial message, I was put directly in touch with the Chief Business Officer.
- YuJa released a security patch after ~2 days, closing the loop-hole for good.

• Webhooky - Feb 2018 thru Sep 2019

- <https://github.com/Windows81/Playing-Webhooky>
- Discord's webhook API allows developers to easily send custom messages from live Roblox games to a shared space on Discord, where data persistence is more accessible than on Roblox's servers.
- In 2018, when Discord abruptly blocked all webhook activity from Roblox servers, I built a lightweight proxy to suit that need.
- Made use of PHP, SQL (MySQL), and PHPMyAdmin to process and store request logs.
- Served over 200 distinct games, including from major studios such as ROLVe.
- Monitored and tracked infrastructure usage stats on cPanel via 000webhost, avoiding excessive downtime.

• Tubeup - Aug 2023

- I use a variety of tools, most of which I install from GitHub, to automate tasks that would otherwise take me months to complete.
- However, some of these tools are missing features or contain bugs which are trivial to fix.
- <https://github.com/bibanon/tubeup>
- Tubeup is a Python utility with 478 stars and new versions being released as of February 2026.
- It uses the yt-dlp and internetarchive libraries to download videos from YouTube to re-upload to the Internet Archive.
- I discovered Tubeup in 2023 to help mitigate the risk of my prior work on YouTube being undiscoverable due to shifting policies.
- On a batch of 234 videos, the program crashed on attempting to download deleted or invalid media.
- I patched Tubeup to alleviate the issue, then I submitted my changes via a pull request on the official project page.
- Unlike most other projects to which I contributed, the author quickly accepted my changes.